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CN5190 Hydrogen Energy and Technology

Essay Topic: Ammonia Cracking for Hydrogen Generation

Under what technological and economic conditions can ammonia cracking provide an efficient, low-carbon and commercially viable hydrogen supply pathway?

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1. Abstract

Singapore generates approximately 95% of its electricity from imported natural gas, while its National Hydrogen Strategy identifies low-carbon hydrogen as a potential source of up to half its power needs by 2050. This review evaluates the technological and economic conditions under which ammonia cracking can provide efficient, low-carbon and commercially viable hydrogen, drawing on mechanistic, catalytic, reactor, techno-economic and life-cycle studies. NH_3 conversion above 99% and process energy efficiencies near 80% are achievable, but ammonia feedstock accounts for 60-65% of hydrogen production costs, while reconversion adds approximately US$2/kg H_2. Cracking is most attractive when end uses require molecular hydrogen, low-carbon ammonia is affordable, and low-temperature catalysts enable effective heat integration, and purification is matched to end-use purity requirements. For Singapore's power sector, direct ammonia combustion was approximately 40% cheaper than cracking-based power generation, so direct combustion or partial cracking suits power generation, while full cracking is better reserved for applications requiring high-purity hydrogen.

2. Motivation and Introduction

Singapore's reliance on imported natural gas, which supplies approximately 95% of its electricity, motivates the transition towards alternative low-carbon fuels [1]. According to the National Hydrogen Strategy announced in 2022, low-carbon hydrogen could supply up to half of Singapore's power needs by 2050 [2]. Given its limited land and renewable energy resources, Singapore could import ammonia as a hydrogen carrier, with its green ammonia market projected to grow from 60 tonnes in 2021 to 18,000 tonnes by 2030 [1].

Ammonia offers a substantial volumetric hydrogen storage advantage over compressed and liquid hydrogen (Fig. 1). Unlike liquid hydrogen, ammonia can be stored at moderate pressure near ambient temperature or refrigerated at atmospheric pressure. Its established global trade, reaching approximately 20 Mt annually, further supports its potential as a hydrogen carrier [5]. Despite the energy losses associated with ammonia synthesis and cracking, long-distance transport can remain economically attractive: a Korean case study estimated delivered hydrogen costs of US$3.5-5.5/kg via ammonia, compared with US$8-12/kg for hydrogen transport, delivery and dispensing in 2017 [6]. However, ammonia's toxicity, material compatibility concerns and combustion-related NO_x and N_2O emissions present significant challenges [6].

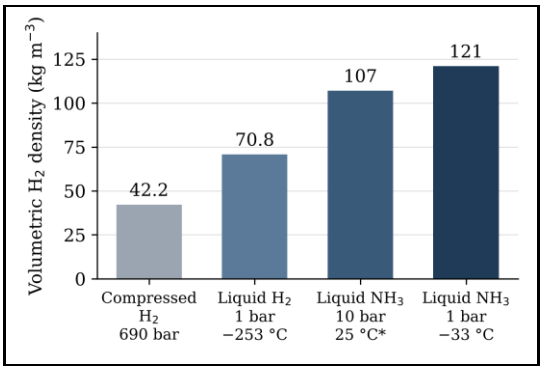


Fig. 1. Volumetric hydrogen storage density of compressed H_2, liquid H_2 and liquid NH_3 under different storage conditions. Data from Cechetto et al. [3] and Makepeace et al. [4].

Ammonia cracking must therefore compete with direct ammonia utilisation. Recovering molecular hydrogen is justified primarily when the end use requires it, such as fuel cells and certain industrial processes, rather than applications where ammonia can be used directly for heat or power. Historically, cracking has attracted limited commercial interest because over 80% of ammonia production serves fertiliser markets, while hydrogen from natural-gas reforming with carbon capture costs approximately US$1.2-2.0/kg [6]. Interest has renewed as hydrogen import strategies increasingly favour ammonia as a transport carrier. Strategy& recommends developing cracking capabilities at destination ports [7], while large-scale facilities are being considered at established ammonia hubs such as Rotterdam, Antwerp and Ulsan [5]. Nevertheless, no publicly documented single-train cracker has demonstrated hydrogen production at hundreds of tonnes per day [3].

3. Scientific Fundamentals

Ammonia decomposition is an endothermic, entropy-driven reaction, with two moles of gaseous reactant producing four moles of gaseous products:

$$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g), \quad \Delta H°_{298} = +91.8 \text{ kJ } (45.9 \text{ kJ per mol } NH_3) \tag{1}$$

The reaction enthalpy of 91.8 kJ per 2 mol NH_3, equivalent to 30.6 kJ per mol H_2 [3], represents approximately 13% of hydrogen's lower heating value (LHV) of 120 MJ kg^{-1}. However, overall cracking processes achieve energy efficiencies of only 79-81% [8], reflecting additional energy demands for feed vaporisation, heating and product separation. High temperatures and low pressures favour ammonia decomposition: at 1 bar, equilibrium conversion exceeds 99% near 400 °C, whereas approximately 600 °C is required at 10 bar [3]. Consequently, pressurised cracking reduces downstream compression requirements at the expense of thermodynamic conversion.

On metal catalysts, ammonia decomposes through adsorption, successive N-H bond cleavage ($NH_3^* \rightarrow NH_2^* \rightarrow NH^* \rightarrow N^*$), and recombinative desorption of N_2 and H_2 [9]. The rate-determining step depends on the metal and reaction conditions: N-H bond cleavage has been identified as limiting on Ru, Rh, Ir, Pt and Pd, whereas N_2 desorption can dominate on Fe, Co and Ni [9].

Nitrogen binding energy serves as a descriptor of catalytic activity. Strong nitrogen binding facilitates N-H bond cleavage but hinders N_2 desorption, whereas weak binding has the opposite effect, producing a Sabatier volcano with Ru near the activity maximum [9] (Fig. 2). Boisen et al. [10] demonstrated that the optimal catalyst for ammonia decomposition differs from that for ammonia synthesis, and that the activity maximum shifts towards stronger nitrogen binding as NH_3 is depleted and H_2 accumulates. Hydrogen also inhibits decomposition: over Ru/CaO, the reaction order in H_2 is −1.9, compared with −1.2 following K promotion [11], as adsorbed hydrogen blocks active sites. Consequently, in situ H_2 removal (Section 5) can enhance reaction kinetics while shifting the equilibrium towards decomposition.

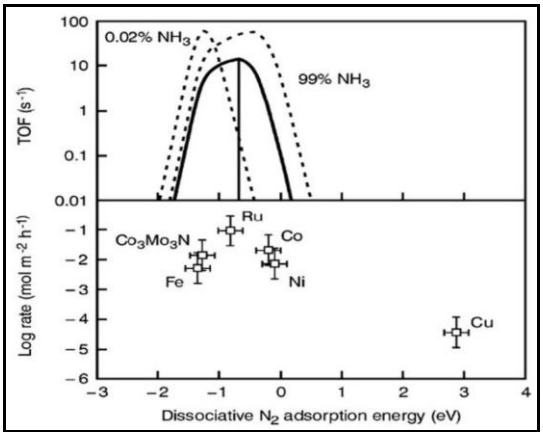


Fig. 2. Calculated turnover frequencies (top) and experimental ammonia decomposition rates (bottom) versus dissociative N_2 adsorption energy. Reproduced from Boisen et al. [10]. Ru lies near the activity maximum, which shifts towards stronger nitrogen binding as NH_3 concentration decreases.

In conventional reactors reviewed by Cechetto et al. [3], NH_3 conversion at 350-400 °C reached only 8-65%, compared with equilibrium conversions of 92-99%. This demonstrates that low-temperature ammonia cracking is primarily limited by reaction kinetics rather than thermodynamics.

4. State-of-the-Art Catalytic Developments

Ruthenium. Ruthenium remains a benchmark catalyst for ammonia decomposition because of its high activity, although its cost and limited abundance constrain commercial deployment [9]. Promoters and supports strongly influence its performance. For an optimised 3 wt% Ru-K/CaO catalyst, potassium promotion reduced the apparent activation energy from 96 to 75 kJ mol^{-1} [11]. At 400 °C, ammonia conversion varied from 30.8% on carbon nanotubes to 53.7% on CaO under comparable Ru and K loadings [11], demonstrating the importance of support selection. At 40 bar and 550 °C, conversion reached approximately 80%, while hydrogen productivity was about 40 times higher than at 1 bar [11], highlighting the trade-off between equilibrium conversion and pressurised hydrogen production.

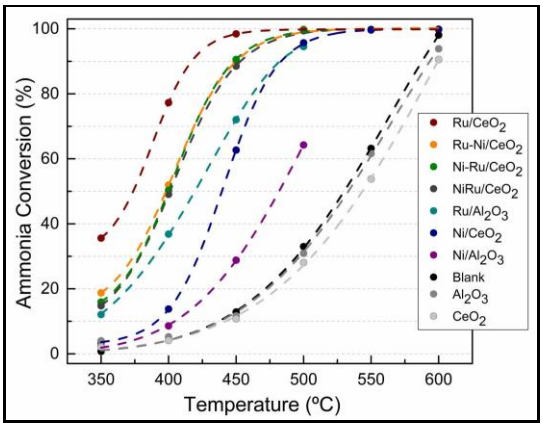


Fig. 3. NH_3 conversion versus temperature over Ni, Ru and Ni-Ru catalysts supported on CeO_2 and Al_2O_3 (1 atm, GHSV = 3,067 h^{-1}, Ar:NH_3 = 1.3:1). Reproduced from Lucentini et al. [12]. Ru/CeO_2 exhibits the highest low-temperature activity.

Supports. Catalyst supports can be as influential as the active metal (Fig. 3). Under identical conditions, Ru/CeO_2 achieved 98% NH_3 conversion at 450 °C, compared with 63% for Ni/CeO_2, while ceria outperformed alumina for both metals [12]. At 400 °C, Ru turnover frequency was 1.8 s^{-1} on CeO_2 versus 1.0 s^{-1} on Al_2O_3, attributed to ceria's redox properties and metal-support interactions rather than acidity [12]. Adding Ni to Ru provided no further improvement, regardless of the order of metal incorporation [12].

Nickel and cobalt. Non-noble catalysts are generally less expensive but require higher operating temperatures [6], making durability an important consideration. During 100 h tests at 450 °C, Ni/CeO_2 maintained its activity, whereas Ru/Al_2O_3 deactivated substantially as Ru particles sintered from approximately 8 to 40 nm, allowing Ni/CeO_2 to outperform it after 85 h [12]. Ru/CeO_2 deactivated more gradually, reportedly through Ru volatilisation [12]. Cobalt lies near Ni on the activity volcano (Fig. 2), with some Co-based formulations exhibiting activity comparable to or exceeding Ru and Ni under particular conditions [9].

Comparing results. Catalyst performance cannot be assessed from conversion alone, as reported values depend on reaction conditions, space velocity and the performance metric used. The absence of standardised testing protocols further complicates direct comparisons [6]. Overall, promoted Ru on basic or reducible oxides is advantageous when low-temperature activity is critical, whereas supported Ni offers a more economical and

established option at 550-750 °C. Co-based catalysts represent a promising non-noble alternative, although their long-term performance and scalability require further validation. Importantly, catalyst price does not necessarily determine hydrogen production cost: despite costing approximately 16 times more per kilogram than Ni (US$542 versus US$33.4/kg), the Ru-based process achieved lower overall costs [8], primarily through reduced operating temperatures and heating requirements.

5. Reactor Technologies and System Integration

Packed-bed crackers. Furnace-heated tubular packed-bed reactors followed by ammonia removal and pressure-swing adsorption (PSA) represent a conventional approach to hydrogen production from ammonia (Fig. 4). In Johnson Matthey's ADEPT process, liquid NH_3 is pumped, vaporised and preheated before catalytic decomposition in externally heated tubes. The product gas is subsequently cooled through heat recovery, treated to remove residual NH_3 and purified by PSA [5]. The PSA tail gas, containing N_2, unrecovered H_2 and residual NH_3, can be recycled as furnace fuel to reduce external energy demand [5]. However, the illustrated configuration also includes natural gas as supplementary fuel. Hydrogen purity must meet end-use requirements: fully cracked ammonia produces a gas mixture containing 75 mol% H_2 and 25 mol% N_2, while ISO 14687 specifies an NH_3 limit of 0.1 ppm for proton-exchange membrane (PEM) fuel-cell applications [3].

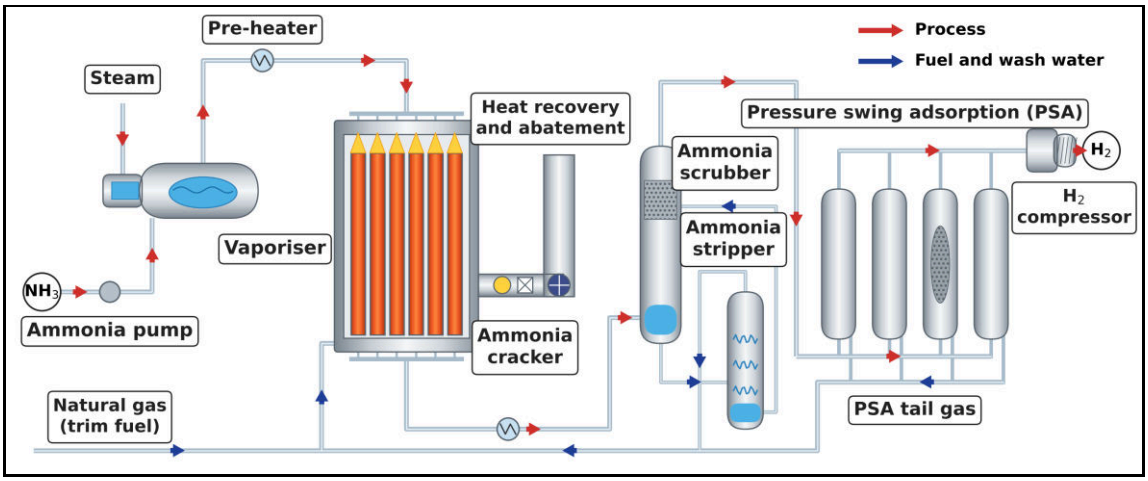


Fig. 4. Johnson Matthey ADEPT ammonia cracking process showing catalytic decomposition, heat recovery, ammonia removal, PSA purification and tail-gas recycling. Redrawn from Mistry [5].

PSA recovery. Hydrogen recovery is critical because ammonia feedstock dominates production costs. Deka et al. [6] reported a Ru-K/CaO process with PSA that achieved 81% H_2 recovery at 96% NH_3 conversion and 83% energy efficiency. Another techno-economic study assumed 75.6% PSA recovery at 99.11 mol% H_2 purity, requiring 7.6-8.0 kg NH_3 per kg H_2, approximately 35-42% above the stoichiometric requirement of 5.6 kg [8]. However, unrecovered H_2 can be combusted as furnace fuel rather than discarded, highlighting the need to optimise hydrogen recovery and heat integration together.

Membrane reactors. Pd-alloy membrane reactors enhance ammonia decomposition by selectively removing H_2 from the catalyst bed, shifting equilibrium towards the products while reducing hydrogen inhibition. At 4 bar and 400 °C, a conventional Ru/Al_2O_3 reactor achieved 65.4% NH_3 conversion against an equilibrium value of 96.7%, whereas a Pd-Ag membrane reactor with vacuum permeation reached 99.3% conversion and 93.5% H_2 recovery [3]. Conversion measures the fraction of NH_3 decomposed, while recovery measures the fraction of generated H_2 collected. Assuming complete conversion, this recovery corresponds to approximately 6.0 kg NH_3 per kg H_2, compared with 7.6-8.0 kg in the PSA-based processes discussed above, although the configurations and heating requirements differ. Pd-Ag alloys mitigate pure Pd's embrittlement below 300 °C and 2 MPa, but membranes remain costly, and plant-scale techno-economic comparisons with packed beds are still lacking [3].

Heat and pressure. Lower cracking temperatures facilitate heat integration and reduce supplementary fuel demand. Cho et al. [8] found that a Ru reactor operating at 567 °C required no additional fuel, whereas a Ni reactor at 736 °C consumed supplementary NH_3. Membrane reactors operating below 400 °C could potentially recover heat from downstream hydrogen users [3]. Reactor pressure introduces another trade-off: increasing pressure from 1.2 to 10 bar reduced unit capital costs by limiting downstream compression, but further pressurisation increased equipment costs and NO_x emissions, so the lowest-cost design operated at the 10-bar ammonia supply pressure [8]. Overall, heat-integrated packed-bed reactors with PSA offer the most established near-term option, while membrane reactors remain promising for high-purity hydrogen production at smaller scales.

6. Techno-Economic and Environmental Assessment

Cost boundaries. Reported hydrogen costs vary substantially because studies adopt different system boundaries. Reconversion cost (REC) excludes ammonia purchase, whereas the levelised cost of hydrogen (LCOH) includes feedstock costs; landed costs additionally account for transportation and distribution. Cho et al. [8] estimated REC values of US$1.86-2.00/kg H_2 for Ru- and Ni-based processes, comparable to the US$2.0/kg H_2 reconversion cost estimated by Strategy& [7] for Gulf-to-Europe hydrogen delivery in 2020 (Fig. 5). In that scenario, reconversion was the largest cost component after hydrogen production, while ammonia conversion and reconversion together accounted for approximately 57% of the US$5.5/kg H_2 landed cost, while shipping and transmission added only about 5% [7].

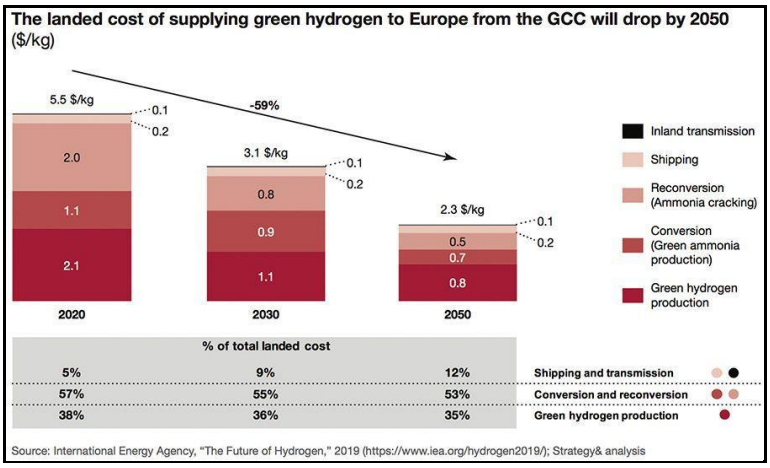


Fig. 5. Projected landed cost of green hydrogen transported as ammonia from the GCC to Europe (2020-2050). Reproduced from Strategy& [7], based on IEA (2019) data.

Feedstock and temperature. Ammonia feedstock dominates hydrogen production costs, accounting for 61-65% of LCOH, while total operating costs represent 86-88% [8]. In a centralised plant, varying ammonia prices by ±75% shifted hydrogen costs from approximately US$3.60 to US$7.40/kg H_2, compared with US$4.50-6.50/kg H_2 across the electricity-price range [6]. At renewable ammonia prices exceeding US$800/t [8], feedstock alone would cost approximately US$6/kg H_2, assuming 7.64 kg NH_3 per kg H_2 (our estimate). Reactor temperature also matters more than catalyst purchase price: the Ru-based process achieved a 7.0% lower REC than the Ni-based process because operating at 567 °C rather than 736 °C eliminated supplementary ammonia fuel demand, while reducing untreated flue-gas NO_x emissions from 1,517 to 863 mg Nm^{-3} [8]. Economies of scale further improve competitiveness, with hydrogen costs falling from US$5.21 to US$3.58/kg H_2 as ammonia feed increased from 25 to 1,000 kmol h^{-1} [6].

Singapore. For a 60.1 MW Singapore power plant fuelled by imported ammonia, Halim et al. [1] estimated LCOH values of US$15.1/kg H_2 for green ammonia cracking and US$10.1/kg H_2 for blue ammonia cracking. Direct ammonia combustion was approximately 40% cheaper than the cracking-based power-generation pathway, primarily by avoiding additional cracking equipment and operating costs [1]. Ammonia prices remained the dominant economic driver. Commercial turbines such as the Mitsubishi H-25 can operate on NH_3-H_2 mixtures containing 30-60 vol% H_2 [1], suggesting that partial cracking without hydrogen purification could offer an intermediate pathway between direct combustion and full cracking.

Environment. The environmental performance of ammonia cracking depends strongly on ammonia production pathways. Approximately 76% of emissions from conventional ammonia production arise from hydrogen generation [6]. For Singapore, green ammonia offers substantially lower life-cycle emissions than conventional ammonia, whether used directly or cracked, whereas coal-derived ammonia could produce emissions exceeding those of the current electricity grid [1]. Blue ammonia may provide a transitional option, although its benefits depend on carbon capture performance and upstream methane emissions. Low-carbon ammonia imports would therefore require credible emissions certification. Local pollutants also remain a concern: direct ammonia combustion can generate NO_x and N_2O [6], while combustion-based cracking shifts NO_x formation to the furnace, potentially requiring exhaust after-treatment [8]. Mistry [5] reports that using ammonia and PSA tail gas instead of natural gas as furnace fuel can achieve emissions below 0.1 kg CO_2e per kg H_2.

7. Challenges, Implementation and Future Developments

Large-scale ammonia cracking remains commercially immature, particularly when high-purity hydrogen production is required. Integrated cracking and purification systems have been assessed at technology readiness levels (TRLs) of 4-6, compared with TRL 9 for conventional fossil-based reforming [6]. Table 1 summarises the principal barriers to commercial deployment, their underlying causes and potential solutions.

Table 1. Key challenges for commercial ammonia cracking.

Challenge	Cause and evidence	Proposed solutions and remaining limitations
Energy penalty	Endothermic cracking at 550-750 °C; process efficiency of 79-81% [8]; supplementary NH_3 combustion generates NO_x [8].	Lower-temperature Ru catalysts, tail-gas heat recovery [5], [8] and membrane reactors near 400 °C [3]. Heat availability and integration remain limiting.
Catalyst cost and durability	Ru is costly and scarce [9]; sintering on Al_2O_3 and reported volatilisation on CeO_2 [12].	K promotion, suitable supports [11], [12] and stable Ni catalysts [12]. Long-duration testing and standardised protocols remain limited [6].
Purity and recovery	Cracked gas contains 25 mol% N_2; PEM applications require NH_3 below 0.1 ppm [3]; PSA recovery of 76-81% [6], [8].	PSA with tail-gas utilisation [5] or Pd-Ag membrane separation [3]. Hydrogen recovery, Pd cost and membrane durability remain concerns.
Scale and siting	No publicly documented single-train plant producing hundreds of tonnes of H_2 daily [3]; NH_3 toxicity creates siting constraints [6].	Co-location with ammonia import and storage infrastructure [5]. Singapore's storage capacity and industrial siting requirements remain constraints [1].

Addressing these challenges requires standardised catalyst and reactor benchmarking, alongside longer-duration stability testing [6]. Machine learning combined with high-throughput experimentation could accelerate catalyst discovery [6], while comparative techno-economic assessments of membrane and conventional reactors across plant capacities remain necessary [3]. For Singapore, ammonia toxicity, limited storage capacity and import dependence favour locating cracking facilities near existing industrial hydrogen users on Jurong Island, where ammonia infrastructure and potential integration with power generation could support deployment [1].

8. Conclusion

Ammonia cracking can provide efficient, low-carbon and commercially viable hydrogen when four conditions are met. Firstly, the end use must require molecular hydrogen, such as fuel cells and industrial applications, rather than heat or power that ammonia can supply directly. Secondly, ammonia must be affordable and produced through low-carbon pathways, as feedstock accounts for 60-65% of hydrogen production costs and upstream emissions strongly influence its carbon footprint. Thirdly, low-temperature catalysts, such as promoted Ru or emerging Co-based formulations, must be combined with effective heat integration to minimise supplementary fuel consumption, operating costs and NO_x emissions. Fourthly, hydrogen purification must balance end-use purity requirements against recovery losses, using PSA for bulk hydrogen or membrane reactors for high-purity applications. For Singapore's power sector, direct ammonia combustion or partial cracking for turbine blends may be more economical than full cracking, which is better suited to applications requiring high-purity hydrogen. Future commercialisation will depend on durable, affordable non-noble catalysts active below 500 °C, lower-cost Pd-lean membranes and Singapore-specific techno-economic assessments using consistent system boundaries.

9. References

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